

4 (a) (i) $(\theta =) \omega t$ (allow any subject if all terms given) B1 [1]

(ii) $(SQ =) r \sin \omega t$ (allow any subject if all terms given) B1 [1]

(b) this is the solution of the equation $a = -\omega^2 x$ M1
 $a = -\omega^2 x$ is the (defining) equation of s.h.m. A1 [2]

(c) (i) $f = \omega / 2\pi$ C1
 $= 4.7 / 2\pi$
 $= 0.75 \text{ Hz}$ A1 [2]

(ii) $v = r\omega$ (r must be identified) C1
 $= 4.7 \times 12$
 $= 56 \text{ cm s}^{-1}$ A1 [2]

5 (a) (i) ratio of charge (on body) and its potential B1 [1]